Evaluating Personality Expression in Large Language Models

Author1, Author2, Author3

University of Zagreb, Faculty of Electrical Engineering and Computing
Unska 3, 10000 Zagreb, Croatia
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Abstract

Large language models are increasingly used to produce text on behalf of users or synthetic personas, yet it is unclear whether they can faithfully express personality in unconstrained writing. On the Pennebaker & King stream-of-consciousness Essays dataset—where human OCEAN profiles (Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism) were established with standard psychological tests—we train a multi-label RoBERTa probe to score trait expression in text, generate essays with GPT-4o-mini under single-trait (Style B) and full-profile (Style A) prompting, and compare predicted OCEAN distributions and per-essay alignment against human baselines. We further analyse linguistic cues via LIWC-style features. This paper describes the end-to-end system: data preparation, probe training, async LLM generation, distributional evaluation, and qualitative error analysis.

1. Introduction

Large language models can produce fluent stream-of-consciousness text, but it remains unclear whether they can reliably adopt a requested personality the way people do when they write freely. In this work we ask a concrete version of that question: if we instruct GPT-4o-mini to write as someone who is, for example, highly extraverted or very anxious, does the resulting essay look—to an automatic personality scorer trained on human writing—like a human essay with that trait, or does the model only mimic obvious wording from the prompt? We study this on the Pennebaker & King Essays dataset, where students wrote unstructured essays and received binary Big Five labels, and we train RoBERTa-base as a multi-label *probe* that predicts those five traits from text; the probe is our measurement instrument, not a claim about ground-truth personality. We then generate new essays in two regimes—Style B varies a single trait at a time (HIGH, LOW, or no trait instruction) to test whether steering one dimension works in isolation, while Style A copies full five-trait profiles from real human annotators to test more realistic multi-trait control—and for every generated essay we run the probe and compare its outputs to the personality we asked for and to the distribution of scores on held-out human essays, using distributional distances and density plots for Style B, per-essay agreement for Style A, and LIWC-style linguistic analysis to inspect which cues differ between human and LLM text. This paper describes that end-to-end pipeline—data preparation, probe training, LLM generation, evaluation, and error analysis—and reports how faithfully explicit personality instructions translate into detectable stylistic shifts. Figure 1 gives a high-level view of the three-phase design; Figure 2 maps the same workflow to the concrete modules in our codebase.

2. Second section

In scientific papers, this section usually (but not necessarily) briefly describes the related research and what makes the presented approach different from it.

2.1. First subsection

This is a subsection of the second section.

2.2. Second subsection

This is the second subsection of the second section. Referencing the (sub)sections in text is performed as follows: "in Section 2.1. we have shown ...".

2.2.1. Sub-subsection example

This is a sub-subsection. If possible, it is better to avoid sub-subsections.

3. Extent of the paper

The paper should have a minimum of 3 and a maximum of 4 pages, plus an additional page for references.

4. Figures and tables

4.1. Figures

Figures are included in LaTeX code immediately *after* the text in which they are first referenced. Our architecture diagrams appear in the Introduction as Figure 1 and Figure 2. Use tilde () to prevent separation between the word "Figure" and its enumeration.

4.2. Tables

There are two types of tables: narrow tables that fit into one column and a wide table that spreads over both columns.

4.2.1. Narrow tables

Table 1 is an example of a narrow table. Do not use vertical lines in tables – vertical tables have no effect and they make tables visually less attractive. We recommend using *booktabs* package for nicer tables.

4.3. Wide tables

Table 2 is an example of a wide table that spreads across both columns. The same can be done for wide figures that should spread across the whole width of the page.

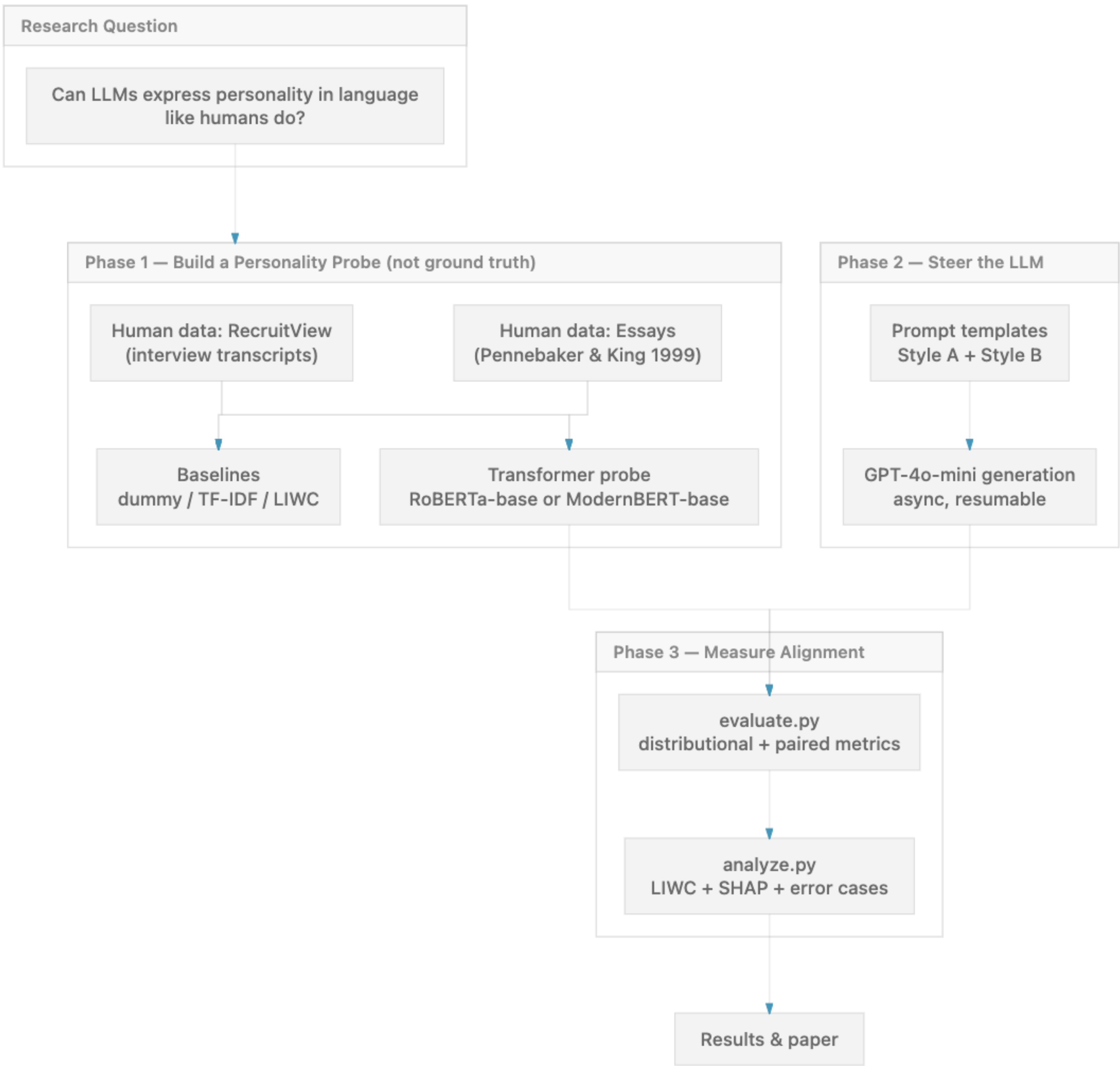

Figure 1: High-level experimental architecture. Phase 1 trains a transformer-based personality *probe* on human text (Pennebaker Essays and RecruitView interview transcripts), alongside simpler baselines. Phase 2 uses Style A and Style B prompt templates to steer GPT-4o-mini generation. Phase 3 scores LLM outputs with the probe and measures alignment via distributional and paired metrics, followed by LIWC-style and SHAP-based linguistic analysis.

Table 1: This is the caption of the table. Table captions should be placed *above* the table.

Heading1	Heading2
One	First row text
Two	Second row text
Three	Third row text
	Fourth row text

5. Math expressions and formulas

Math expressions and formulas that appear within the sentence should be written inside the so-called *inline* math environment: $2+3$, $\sqrt{16}$, $h(x) = \mathbf{1}(\theta_1 x_1 + \theta_0 > 0)$. Larger expressions and formulas (e.g., equations) should be written in the so-called *displayed* math environment:

$$b_k^{(i)} = \begin{cases} 1 & \text{if } k = \text{argmin}_j \|\mathbf{x}^{(i)} - \mu_j\|, \\ 0 & \text{otherwise} \end{cases}$$

Math expressions which you reference in the text should be written inside the *equation* environment:

$$J = \sum_{i=1}^{N} \sum_{k=1}^{K} b_k^{(i)} \|\mathbf{x}^{(i)} - \mu_k\|^2 \tag{1}$$

Now you can reference equation (1). If the paragraph

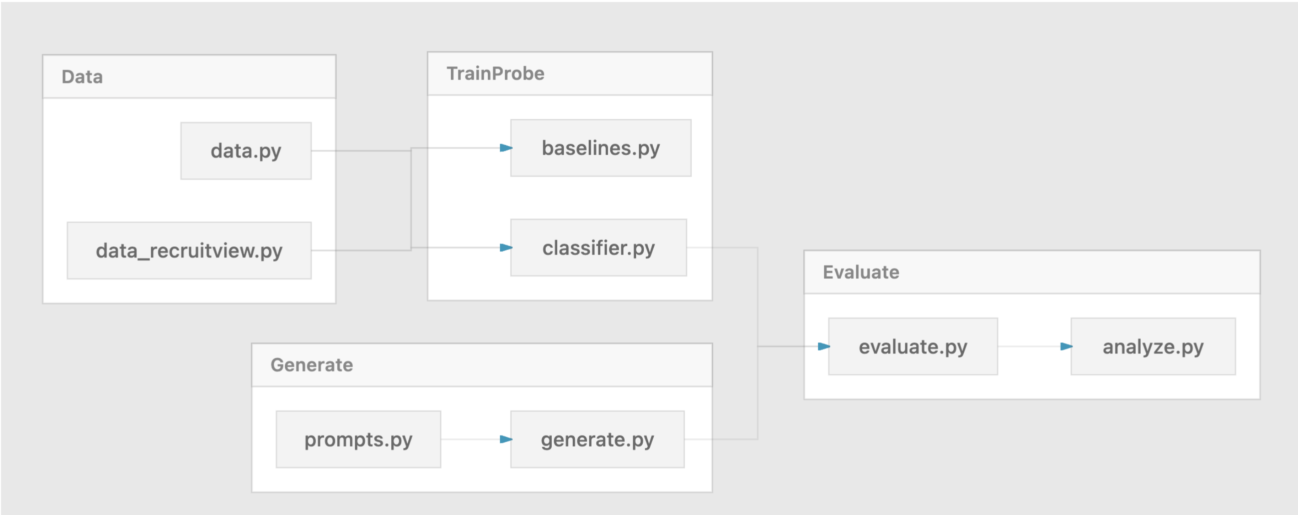

Figure 2: Code pipeline overview. Human data are prepared by and , then used to train baselines and the probe (,). Prompt templates () drive async LLM generation (). Probe predictions and generated text are evaluated () and analysed linguistically ().

Table 2: Wide-table caption

Heading1	Heading2	Heading3
A	A very long text, longer that the width of a single column	128
B	A very long text, longer that the width of a single column	3123
C	A very long text, longer that the width of a single column	-32

continues right after the formula

$$f(x) = x^2 + \varepsilon \tag{2}$$

like this one does, use the command *noindent* after the equation to remove the indentation of the row.

Multi-letter words in the math environment should be written inside the command *mathit*, otherwise LaTeX will insert spacing between the letters to denote the multiplication of values denoted by symbols. For example, compare $Consistent(h, \mathcal{D})$ and
$\mathit{Consistent}(h, \mathcal{D})$.

If you need a math symbol, but you don't know the corresponding LaTeX command that generates it, try *Detexify*.[1]

6. Referencing literature

References to other publications should be written in brackets with the last name of the first author and the year of publication, e.g., (**?**). Multiple references are written in sequence, one after another, separated by semicolon and without whitespaces in between, e.g., (**?**; **?**; **?**). References are typically written at the end of the sentence and necessarily before the sentence punctuation.

If the publication is authored by more than one author, only the name of the first author is written, after which abbreviation *et al.*, meaning *et alia*, i.e., and others is written as in (**?**). If the publication is authored by only two authors, then the last names of both authors are written (**?**).

If the name of the author is incorporated into the text of the sentence, it should not be in the brackets (only the year should be there). E.g., "**?**) suggested that …". The difference is whether you reference the publication or the author who wrote it.

The list of all literature references is given alphabetically at the end of the paper. The form of the reference depends on the type of the bibliographic unit: conference papers, (**?**), books (**?**), journal articles (**?**), doctoral dissertations (**?**), and book chapters (**?**).

All of this is automatically produced when using BibTeX. Insert all the BibTeX entries into the file , and then reference them via their symbolic names.

7. Conclusion

Conclusion is the last enumerated section of the paper. It should not exceed half of a column and is typically split into 2–3 paragraphs. No new information should be presented in the conclusion; this section only summarizes and concludes the paper.

Acknowledgements

If suitable, you can include the *Acknowledgements* section before inserting the literature references in order to thank those who helped you in any way to deliver the paper, but are not co-authors of the paper.

[1]